



OPEN A novel XAI framework for explainable AI-ECG using generative counterfactual XAI (GCX)

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Generative Counterfactual Explainable Artificial Intelligence (XAI) offers a novel approach to understanding how AI models interpret electrocardiograms (ECGs). Traditional explanation methods focus on highlighting important ECG segments but often fail to clarify why these segments matter or how their alteration affects model predictions. In contrast, the proposed framework explores “what-if” scenarios, generating counterfactual ECGs that increase or decrease a model’s predictive values. This approach has the potential to increase clinicians’ trust specific changes—such as increased T wave amplitude or PR interval prolongation—influence the model’s decisions. Through a series of validation experiments, the framework demonstrates its ability to produce counterfactual ECGs that closely align with established clinical knowledge, including characteristic alterations associated with potassium imbalances and atrial fibrillation. By clearly visualizing how incremental modifications in ECG morphology and rhythm affect artificial intelligence-applied ECG (AI-ECG) predictions, this generative counterfactual method moves beyond static attribution maps and has the potential to increase clinicians’ trust in AI-ECG systems. As a result, this approach offers a promising path toward enhancing the explainability and clinical reliability of AI-based tools for cardiovascular diagnostics.

Keywords AI-ECG, Explainable AI, XAI, Generative counterfactual XAI, Deep learning

Artificial intelligence (AI) applied to electrocardiograms (ECGs) (AI-ECG) has demonstrated remarkable proficiency in diagnosing and predicting cardiovascular and other systemic diseases, often equaling or exceeding the expertise of cardiologists^{1–5}. Despite these achievements, deep learning-based AI-ECG models face a critical challenge: their decision-making processes are not transparent and are often regarded as “black boxes”^{6,7}. These limitations are significant concerns for clinicians, who depend on understanding AI decisions to trust and use these technologies effectively in patient care. Thus, the role of Explainable Artificial Intelligence (XAI) in interpreting AI decisions is becoming increasingly crucial⁸. XAI aims to develop and refine AI systems that are not only effective, but also explainable and reliable for clinical use⁹. In this context, ‘explainable’ refers to the ability to understand why the AI produces a specific result, whereas ‘reliable’ pertains to whether the model’s output aligns with clinical knowledge¹⁰.

Attribution-based XAI methods, such as Saliency Maps and Grad-CAM, have been predominantly used to interpret AI-ECG^{11,12}. These techniques generate importance maps by quantifying the contribution of each input signal to AI-ECG predictions, which are typically visualized as heatmaps. Saliency maps may show that the AI-ECG focuses on certain segments, but they fail to clarify whether changes—such as widening, narrowing, flattening, or inversion—are responsible for influencing the model’s decisions. Additionally, these methods do not account for changes in rhythm, which is critical for the diagnosis of arrhythmias. Moreover, one study found that identical saliency maps could be interpreted in opposite ways depending on the viewer’s prior beliefs, revealing a risk of confirmation bias.¹³ Recent studies have demonstrated that, in ECG research, relying exclusively on heatmap-based explanations is inadequate, highlighting the need for a comprehensive approach that incorporates multiple XAI methods^{14,15}. Consequently, the interpretation provided by attribution-based XAI methods alone is often insufficient.

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To address limitations in attribution-based XAI in AI-ECG, we build upon existing Generative Counterfactual-based XAI (GCX) methods, which explore “what-if” scenarios to determine how specific changes in input data could influence predictions^{16,17}. To generate these scenarios, generative AI—a technique capable of synthesizing plausible input data—is used to create CF ECG. GCX not only highlights where the model is focusing, but also visualizes how specific changes in the ECG influence predictions, thereby providing a clear explanation of the model’s decision-making process.

GCX methods have been studied in medical AI, particularly in image-based domains. For example, Mertes et al. demonstrated that counterfactual images improved users’ understanding and trust in AI predictions for pneumonia detection, outperforming traditional saliency map methods¹⁸. Similarly, Singla et al. applied counterfactual explanations to chest X-ray diagnosis, showing that the model’s decision shifts aligned with clinically meaningful features such as cardiothoracic ratio and costophrenic angle, thereby validating the medical plausibility of the explanations¹⁹.

Although GCX methods have been applied in image domains, they have not yet been systematically validated for ECG analysis. Unlike images, ECGs are time-series data with complex rhythmic dependencies, which require domain-specific validation. In this study, we present the first comprehensive validation of a GCX method specifically adapted for AI-ECG models. However, GCX alone is inherently limited to providing single-instance (local) explanations, which restricts its ability to capture broader patterns in model behavior (global). To address this limitation, we propose a structured GCX framework that systematically analyzes multiple CF ECGs to identify consistent and clinically meaningful features.

Methods

Definition of GCX framework

The GCX framework is a structured process to explain AI-ECG model predictions using the GCX (Fig. 1b). It consists of two stages. First, the GCX method is used to generate two types of CF ECGs from the original ECG: Positive and Negative CF ECG. A Positive CF ECG is a modified version of the original that yields a higher prediction from the AI-ECG model, while a Negative CF ECG yields a lower one. For example, in a T wave amplitude regression AI-ECG model, if the original ECG has a predicted value of 0.3 millivolts (mV), GCX generates Positive and Negative CF ECGs with values higher or lower than 0.3 mV, respectively. Second, ECG features are extracted from the original and CF ECGs, and then subjected to statistical analysis. This process offers systematic summarization of how specific ECG features contribute to positive or negative changes in model predictions at the population level.

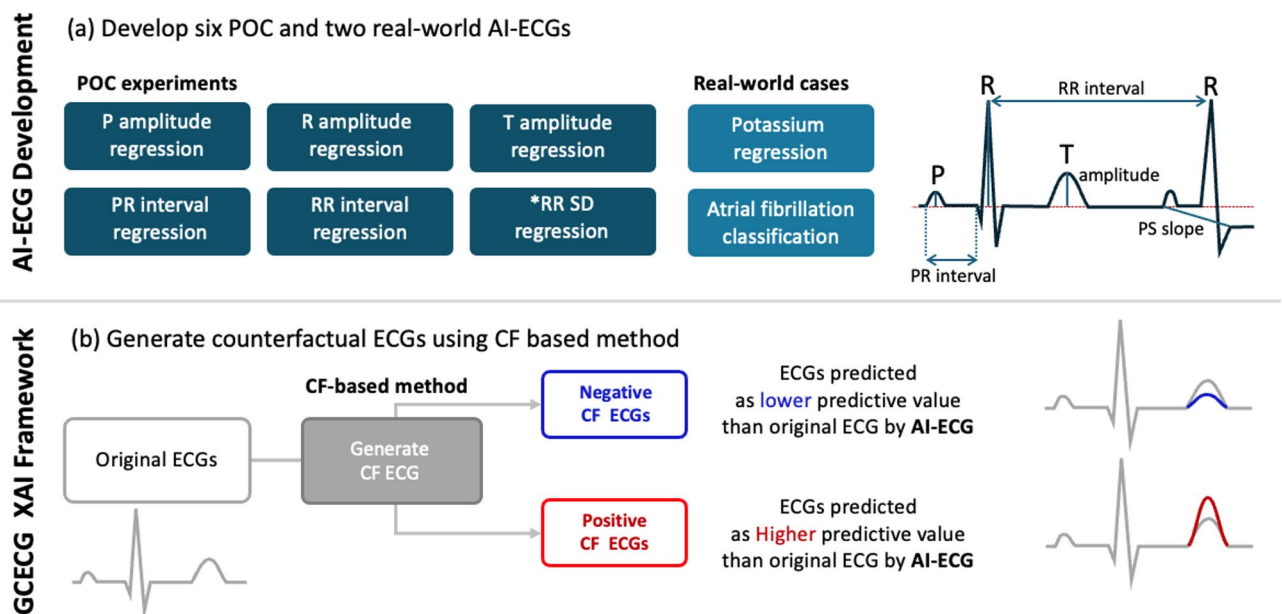


Fig. 1. overview of study flow. **(a)** To assess the effectiveness and utility of Generative Counterfactual-based XAI (GCX), we constructed a total of eight AI-ECGs and aimed to interpret them using GCX. Six ECG feature regression models were interpreted via GCX to verify whether GCX could recognize and interpret the core components of the ECGs. Additionally, we used GCX to interpret the potassium level regression and atrial fibrillation classification models. **(b)** GCX generates modified ECGs reflecting changes in predictive output of specific AI-ECG models, either increasing or decreasing. This approach allows us to evaluate whether the ECGs produced by GCX align with theoretical ECG changes associated with the clinical status. *RR SD; RR interval standard deviation.

Study design

We conducted a proof-of-concept (POC) study to validate the GCX using publicly available clinical ECG datasets (Fig. 1a). As an initial step, GCX was applied to six AI-ECG models trained to regress core morphological and rhythm-related features derived from lead II: P, R, and T wave amplitudes; PR interval; RR interval; and RR interval standard deviation (RR SD). These features were selected due to their quantitative definitions and well-established clinical interpretability, providing a robust foundation to assess whether GCX can generate counterfactual ECGs that induce coherent and physiologically plausible modulations in model output.

Upon establishing the internal validity of GCX, we evaluated its clinical utility by applying the GCX framework to two additional AI-ECG models: one predicting serum potassium concentration (regression task) and another classifying AF (classification task). These downstream tasks were selected due to the well-documented ECG manifestations associated with electrolyte imbalance and arrhythmia. For instance, hyperkalemia is typically reflected in ECG by tall, peaked T waves, prolonged PR intervals, and QRS complex widening. Conversely, hypokalemia is associated with increased P wave amplitude, diffuse ST-segment depression, T wave flattening or inversion, and the appearance of U waves. AF, in contrast, is electrographically defined by the absence of discernible P waves and a characteristically irregular RR interval pattern.

AI-ECG development and data collection

We developed eight AI-ECG models and explain them using the GCX. The AI-ECG models were developed using the ResNet architecture. We used the PTB-XL dataset¹⁸ and the MIMIC-IV ECG dataset¹⁹. Detailed process for developing the AI-ECG models is described in Supplementary Text 1–3.

Generation of CF ECGs using GCX

GCX utilizes StyleGAN2, a generative AI model, to generate CF ECGs for AI-ECG f interpretation²⁰. It generates a CF ECG x'_{pos} from the original ECG x . In this process, GCX ensures that the predictive value $f(x')$ of the Positive CF ECG is higher than the predictive value $f(x)$ of the original ECG. Conversely, the Negative CF ECG x'_{neg} has a lower predictive value than the original. Since the main objective of this study is to validate the GCX, detailed descriptions of the process to generate CF ECGs are provided in Supplementary Text 4.

$$x'_{pos} = GCECG(x)$$

$$f(x) < f(x'_{pos})$$

where x is the original ECG and x'_{pos} is Positive CF ECG and f is ECG AI model.

Visualization of CF ECGs

The median beat method was used to extract representative ECGs for the POC experiments²¹. In POC cases where rhythm changes are crucial, such as with the RR interval and RR SD regression models, we visualized 5-s ECGs. Progressive CF ECG plots were used to validate two AI-ECG models by comparing both Positive and Negative CF ECGs with baseline ECGs. At this stage, saliency maps were visualized alongside the Progressive CF ECG plots for comparative analysis. All visualizations were performed using Python's Matplotlib (v2.1.5)²².

ECG feature analysis and statistics

We conducted an ECG feature analysis using paired t-tests and paired box plots. First, ECG features were extracted from the original ECGs, as well as from positive CF ECGs and negative CF ECGs. A detailed explanation of the feature extraction algorithm is provided in Supplementary Text 2. Then, for each ECG feature, a separate paired t-test was performed to statistically compare the values before and after counterfactual modification—either between the original and Positive CF ECGs or between the original and Negative CF ECGs. These analyses were conducted independently for each feature and comparison. The results of these paired t-tests were presented alongside paired box plots. In these plots, the left box represents the feature values before modification, and the right box represents those after modification. Each paired CF ECG is connected by lines: red solid lines indicate an increase, while blue solid lines indicate a decrease.

For the potassium level regression models, which involved two abnormal states—hyperkalemia and hypokalemia—ECG feature analyses were conducted separately for original ECGs and their corresponding positive CF ECGs (hyperkalemia) and negative CF ECGs (hypokalemia). Specifically, T amplitude, PR interval, and QRS duration were compared for positive CF ECGs, while T amplitude, P amplitude, and PS slope were compared for negative CF ECGs. The PS slope is the slope connecting the onset of the P wave to the offset of the QRS complex; a lower slope indicates a stronger presence of ST depression. For AF classification, the features of negative CF ECGs (non-AF) and positive CF ECGs (AF) were analyzed, focusing on P amplitude and RR SD. Paired t-tests were performed using Python's SciPy library, version (v1.10.1)²³. Statistical significance was assessed using a two-sided threshold of $p < 0.05$.

Results

Table 1 shows baseline characteristics of the datasets used for developing AI-ECG models. The PTB-XL dataset comprises 18,869 patients and 21,799 ECGs, with 55.36% male participants. This dataset includes 1,514 ECGs (6.95%) classified as AF, while the remaining 20,285 ECGs (93.05%) were non-AF. The MIMIC IV dataset includes 104,804 patients and 238,262 ECGs, with 49.06% male participants. This dataset includes 16,964 ECGs (7.12%) identified as hypokalemic (< 3.5 mmol/L) and 14,101 ECGs (5.92%) as hyperkalemic (> 5.5 mmol/L).

In the POC experiments, ECG features—including P, R, and T wave amplitudes and the PR interval—were analyzed using median beat visualizations. The Positive CF ECGs demonstrated increased amplitudes of the P, R,

Dataset name	PTB-XL	MIMIC IV
Patients N	18,869	104,804
ECG N	21,799	238,262
Male (%)	55.36	49.06
Age, mean (^a SD)	59.83 (16.95)	60.54 (17.18)
Heart rate, bpm, mean (SD)	74.17 (16.94)	81.14 (21.65)
QRS duration, ms, mean (SD)	98.70 (16.83)	98.60 (20.80)
QT interval, ms, mean (SD)	387.37 (38.28)	382.24 (48.56)
PR interval, ms, mean (SD)	183.32 (33.29)	157.39 (32.50)
P wave axis, degree, mean (SD)	38.93 (26.17)	40.24 (36.67)
R wave axis, degree, mean (SD)	16.32 (39.93)	11.84 (43.48)
T wave axis, degree, mean (SD)	37.07 (50.22)	36.98 (54.47)
Potassium level, mmol/L, mean (SD)	–	4.30 (0.78)
Atrial fibrillation		
Atrial fibrillation ECGs (%)	1,514 (6.95)	–
Non-atrial fibrillation ECGs (%)	20,285 (93.05)	–
Hypokalemia (< 3.5 mmol/L)		
ECG N (%)	–	16,964 (7.12)
Potassium level, mmol/L, mean (SD)	–	3.19 (0.24)
Hyperkalemia (> 5.5 mmol/L)		
ECG N (%)	–	14,101 (5.92)
Potassium level, mmol/L, mean (SD)	–	6.49 (0.92)

Table1. Baseline Characteristics for ECG datasets: PTB-XL and MIMIC-IV. ^aSD: standard deviation.

and T waves compared to the original ECGs, while the Negative CF ECGs showed decreased amplitudes (Fig. 3). The PR interval panel showed that Positive CF ECGs had a prolonged PR interval, whereas Negative CF ECGs had a shortened PR interval (Fig. 2). For the RR interval and RR SD, the Positive CF ECGs exhibited longer RR intervals and increased variability (RR SD), whereas the Negative CF ECGs exhibited shorter RR intervals and decreased variability (Fig. 2).

For the potassium regression AI-ECG model, Positive CF ECGs indicative of hyperkalemia revealed increased T wave amplitude, prolonged PR interval, and QRS widening, while Negative CF ECGs for hypokalemia show decreased T wave amplitude, increased P wave amplitude, widespread ST depression, and prominent U waves (Fig. 2). The progressive CF plots in Fig. 3 provide a dynamic view of how these features evolve as predicted potassium levels change, offering a continuous and detailed depiction of the morphological changes initially observed in Fig. 2. In contrast, the saliency map emphasizes the T wave amplitude but does not provide additional morphological information.

For the AF classification AI-ECG model, Fig. 2 demonstrates that Positive CF ECGs, associated with higher AF probabilities, are characterized by the absence of the P wave. However, the irregular rhythm is not clearly visible in the median beat visualizations of Fig. 2. This feature becomes apparent in Fig. 3, where progressive CF plots effectively capture not only the absence of the P wave but also the gradual emergence of rhythm irregularities as AF probability increases. Importantly, progressive CF plots also captured the beat-to-beat variability, reflecting the characteristic ‘irregularly irregular’ pattern typical of AF. This dynamic visualization complements the static observations in Fig. 2, providing a more comprehensive understanding of how AF-related changes develop over time. The saliency map, while highlighting the P wave, does not offer additional morphological details and fails to capture the rhythm changes.

All ECG features listed above were statistically verified using paired t-tests, each demonstrating statistically significant outcomes. We analyzed ECG features related to potassium levels and AF, focusing on three specific scenarios: potassium (original to Positive CF ECGs), potassium (original to Negative CF ECGs), and AF (Negative to Positive CF ECGs). Figure 4 presents the results of paired t-tests for each scenario, with detailed values reported in Supplementary Table 1.

For the potassium regression model, when comparing the original ECGs to Positive CF ECGs, there are notable changes that align with clinical hyperkalemia. The Positive CF ECGs showed an increased T wave amplitude, prolonged PR interval, and widened QRS complex. These morphological changes are consistent with typical ECG manifestations of elevated potassium levels in the blood.

In contrast, comparison between the original ECGs and Negative CF ECGs for potassium regression revealed features indicative of hypokalemia. The Negative CF ECGs exhibit a decreased T wave amplitude, increased P wave amplitude, and a reduced PS slope indicating stronger ST depression. These changes are characteristic of ECG alterations associated with hypokalemia.

In the AF detection model, the transition from Negative to Positive CF ECGs highlighted the distinctive features of AF. The Positive CF ECGs are marked by the absence of the P wave and an irregular rhythm, which are hallmark features of AF. This transformation aligns with clinical patterns observed in patients with increasing risk of AF.

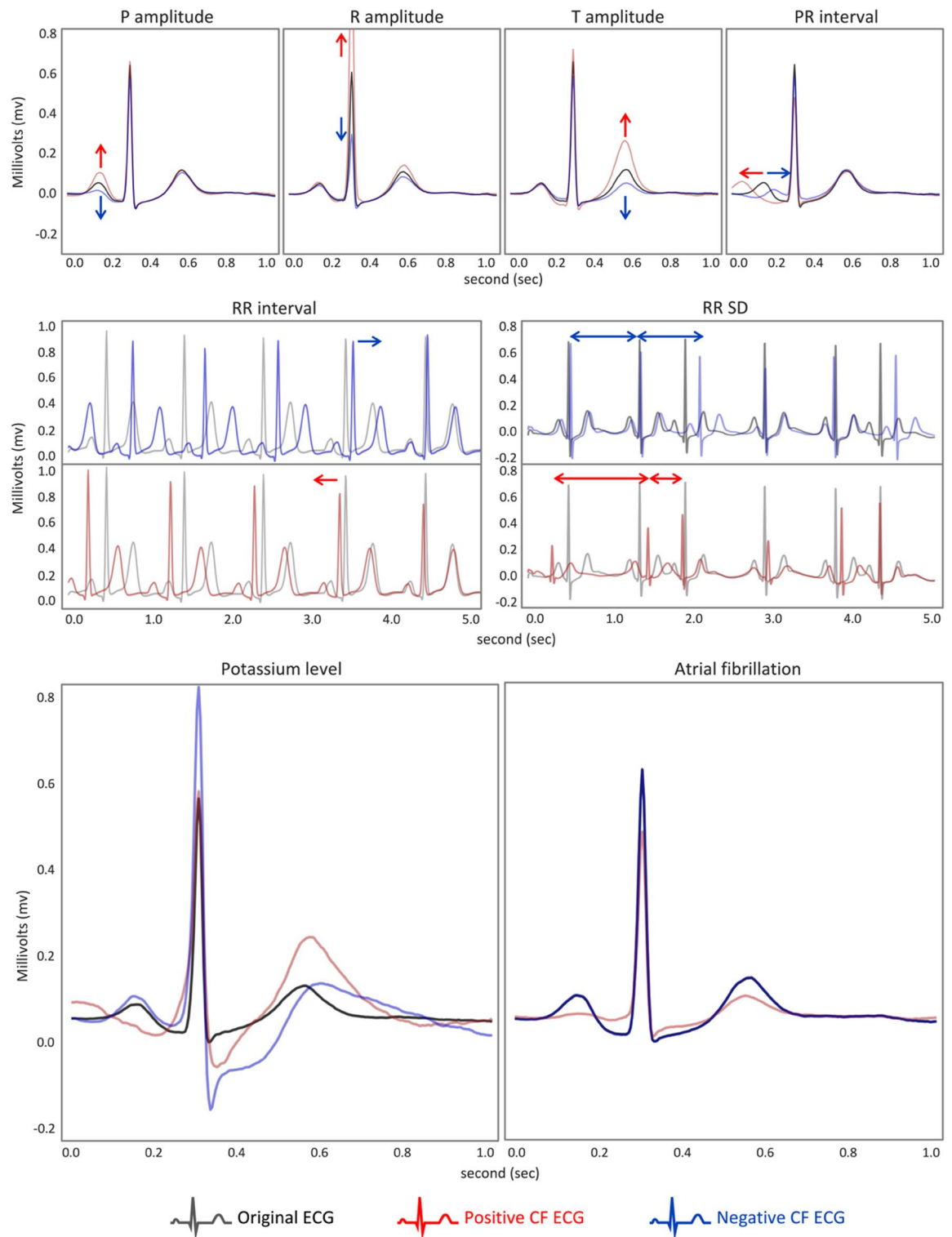


Fig. 2. Visualization of CF ECGs for each AI-ECG model. The figure demonstrates GCX's ability to produce counterfactual ECGs (CF ECGs) and highlights morphological differences between original (black), Positive CF (red), and Negative CF (blue) ECGs. The panels for P, R, and T wave amplitudes and PR interval show median beat visualizations, indicating their significance in each AI-ECG prediction. The RR interval and RR SD panels are 5-s ECG examples of CF ECGs instead of median beat visualizations, as rhythm changes cannot be shown using median beats. The Potassium and Atrial Fibrillation panels also use median beat visualizations to highlight typical changes, such as increased T amplitude in higher potassium levels and the disappearance of the P wave in AF cases.

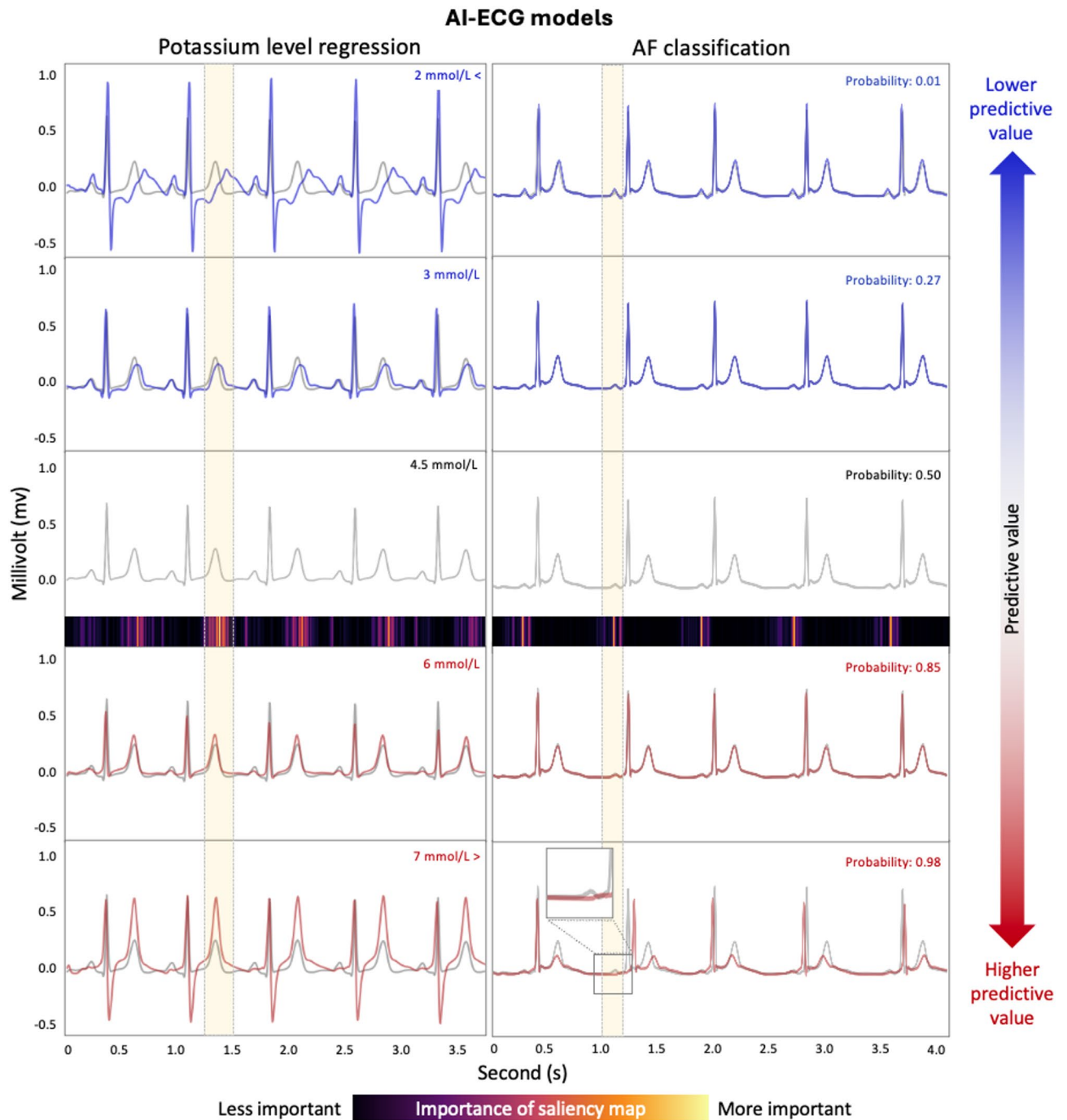


Fig. 3. Progressive CF plots. It demonstrates how ECG morphology changes influence predictive values. The black ECG represents the baseline ECG, while the red and blue ECGs represent Positive and Negative CF ECGs, respectively. It shows the progression of morphological changes as the predictive value varies from the baseline ECG. The colormap shows importance levels of the saliency map, with low-importance segments in black and high-importance segments in yellow. The potassium section shows the progressive changes in ECG morphology corresponding to hyperkalemia and hypokalemia. The ECG traces reflect increasing predictive potassium levels from 2 to 7 mmol/L, with notable changes in T wave amplitude and the shapes of the P and QRS waves. For AF, the figure illustrates clinical features with increasing AF probabilities, ranging from 0.01 to 0.98. The ECG traces show progressive AF characteristics such as rhythm irregularity and the modification or absence of P waves.

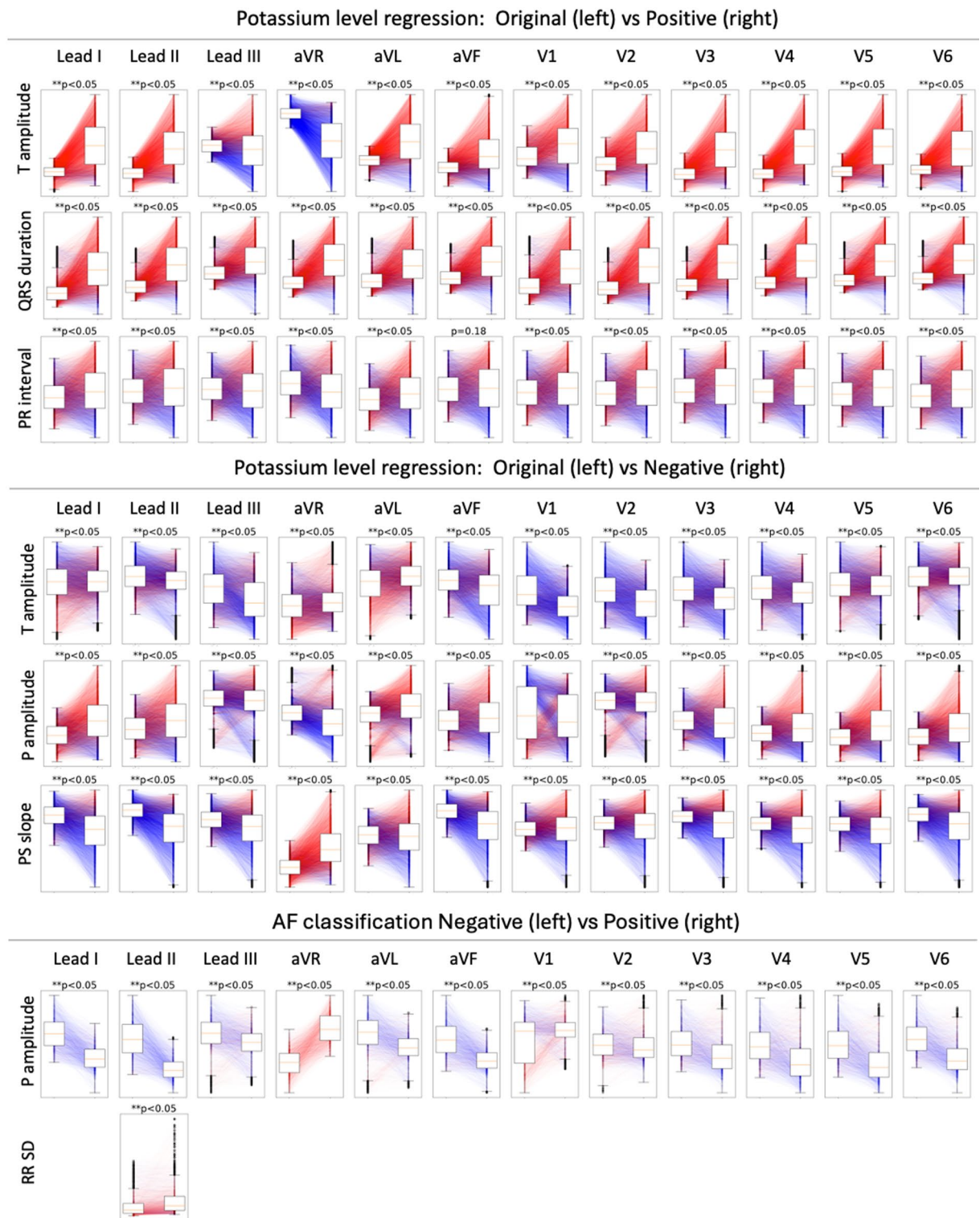


Fig. 4. Paired t-test box plot. It presents paired t-test box plots comparing ECG feature values between original ECGs and CF ECGs generated using the GCX method. The plots illustrate the distributions of various ECG features, such as P amplitude, R amplitude, T amplitude, PR interval, RR interval, RR interval standard deviation (RR SD), and PS slope between paired ECGs. Red lines connect paired features that increased, while blue lines connect those that decreased. The box plots reveal statistically significant changes, demonstrating GCX's ability to modify crucial ECG attributes. The p-values of the paired t-tests are indicated above the plots.

Discussion

This study aimed to validate the GCX framework for explaining AI-ECG models, particularly in the contexts of ECG feature regression, potassium level regression, and AF classification. By leveraging generated CF ECGs, we sought to enhance both the explainability and clinical applicability of AI-driven ECG diagnostics. Although GCX has been partially applied to ECGs in previous studies, to our knowledge, this is the first study to rigorously evaluate a structured GCX framework specifically tailored for AI-ECG explanation^{24,25}.

We first conducted POC experiments to assess whether GCX could recognize and modify six representative ECG features: P wave amplitude, R wave amplitude, T wave amplitude, PR interval, RR interval, and RR SD. The results confirmed that GCX effectively identified and modified these features in a physiologically coherent manner. Building on this foundation, we applied the GCX framework in two clinical tasks: potassium level regression and AF classification. The GCX framework provided explanations and findings that were consistent with established clinical knowledge of ECG changes associated with blood potassium levels and AF diagnosis.

In the potassium level regression model, positive CF ECGs corresponding to higher potassium values showed increased T wave amplitude with peaked morphology, a flattened P wave with prolonged PR interval, and a widened QRS complex (Figs. 2, 3, Supplementary Fig. 1). However, other known hyperkalemia-related ECG changes, such as conduction disturbances or bradyarrhythmia, were not observed using this framework. Negative CF ECGs, indicative of lower potassium levels, displayed decreased T wave amplitude, increased P wave amplitude, a prolonged PR interval, ST depression, and prolonged QT interval with fusion of T and U waves. These findings were consistent with established ECG knowledge.

In the AF classification model, Positive CF ECGs associated with higher AF probabilities exhibited hallmark features such as the disappearance of P waves and the emergence of an ‘irregularly irregular’ rhythm. As shown in Figs. 2, 3 and Supplementary Fig. 2, GCX visualizations demonstrated not only the absence of the P wave but also subtle yet clinically meaningful shifts in the positions of the R peaks, clearly indicating rhythm irregularities that are characteristic of AF. In contrast, traditional attribution-based methods such as saliency maps primarily highlight static features—in this case, the P wave—and fail to capture dynamic rhythm changes or beat-to-beat morphological variability. Thus, GCX provides comprehensive explanations of both morphological and rhythm-related alterations in ECG signals.

GCX can serve not only as a model explanation tool but also as a practical aid for clinical application of AI-ECG. Conventional ECG reports often summarize findings using general terms such as “hyperkalemia” or “tall T waves,” but they provide limited visual or contextual information on specific waveform changes. This lack of clarity can hinder interpretation, particularly for clinicians without advanced ECG training. GCX addresses this gap by generating CF ECGs that allow for visual comparisons between pathological and non-pathological states. As illustrated in Fig. 5, the potassium regression model predicts a potassium level of 6.24 mmol/L, indicating hyperkalemia. A negative CF ECG showing a lower potassium level highlights differences in T wave amplitude and QRS width, helping clinicians understand which features contributed to the AI-ECG’s prediction. Even when the clinician does not initially recall the ECG patterns of hyperkalemia, this side-by-side visualization supports recognition, learning, and more confident interpretation.

The GCX framework may also contribute to scientific discovery by revealing ECG features that are not yet fully understood or described in clinical guidelines. Some diseases, such as heart failure and aortic stenosis, are increasingly detected by AI-ECG models despite lacking well-defined diagnostic criteria in standard ECG interpretation^{26,27}. In these cases, GCX can help visualize the specific waveform patterns that influence model predictions. This creates opportunities to identify novel ECG signatures that may be clinically relevant but currently overlooked. By doing so, GCX shifts from a passive explanatory tool to an active driver of research, potentially guiding new investigations into the electrophysiological characteristics of complex cardiovascular diseases^{26,27}.

While our study demonstrates the utility of GCX, several limitations must be considered. First, CF ECGs may not always reflect clinically accurate patterns. This occurs when the underlying AI-ECG model is trained on biased data or captures unintended artifacts. As a result, GCX may highlight features that are not aligned with the target disease. To ensure reliable explanation, GCX should be applied to AI-ECG models that are thoroughly validated and trained on diverse, high-quality datasets. Second, GCX may occasionally generate physiologically implausible ECGs. This is due to its core mechanism of manipulating input signals to alter model predictions, without explicitly enforcing biological constraints. Such outputs could mislead users or reduce trust in the AI-ECG system. Therefore, CF ECGs should always be reviewed alongside expert clinical judgment, and future improvements should focus on integrating physiological priors or rules into the generative process. Third, the proprietary ECG feature extraction algorithm used in this study limits full reproducibility. External researchers may face challenges in replicating the model pipeline in its entirety. To enhance transparency, we have described the development process of the feature extraction algorithm in detail in Supplementary Texts 1 and 2. Moreover, all ECG feature values extracted using this method and used in model training are provided in Supplementary File 1. Importantly, GCX itself is agnostic to the feature extraction method and can be applied to models built using publicly available, well-validated ECG feature extraction tools, supporting flexible and reproducible implementation in future research.

Conclusion

In conclusion, the GCX framework represents a meaningful advancement in the explainability and reliability of AI-ECG models by providing clear and intuitive explanations for specific ECG features, morphological patterns, and rhythm changes, aligned with established ECG knowledge. By generating and analyzing CF ECGs, GCX effectively addresses key limitations of traditional XAI methodologies for ECG analysis and holds substantial promise for both clinical and technical applications. Future research should aim to extend the use of GCX to a

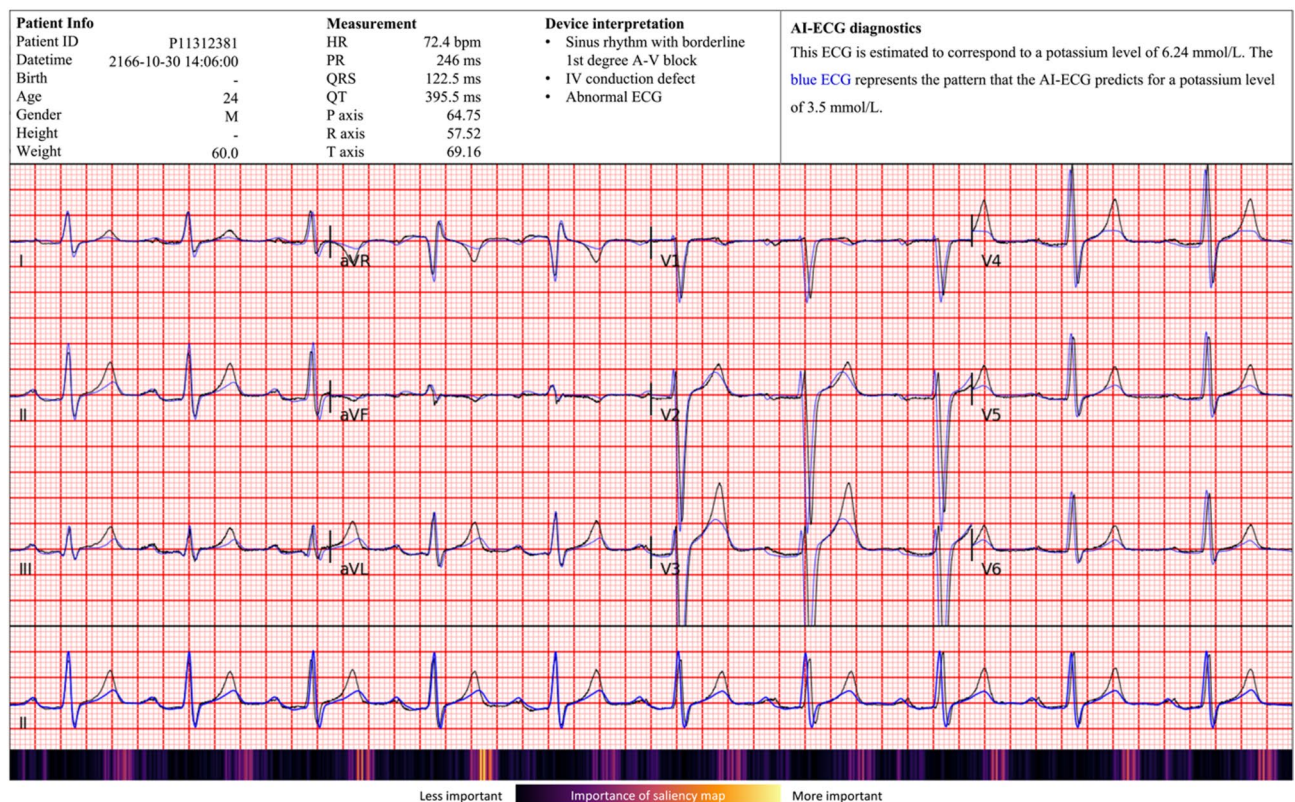


Fig. 5. Example of an ECG report integrated with GCX for potassium level regression model. The patient's actual potassium level was 6.59 mmol/L, and the AI-ECG model predicted a level of 6.24 mmol/L, indicating hyperkalemia. The overlaid blue trace represents the negative CF ECG generated by GCX, simulating a lower potassium level with a predicted value of 3.5 mmol/L. By visually comparing the original and CF ECGs, clinicians can clearly identify waveform features—such as reduced T wave amplitude and narrower QRS complexes—that contributed to the AI-ECG model's prediction.

broader range of clinical tasks and AI-ECG models to further validate its effectiveness and utility. Additionally, evaluating its performance across diverse populations and ECG datasets will be critical to improving its generalizability.

Data availability

All datasets used in this study are publicly available. The PTB-XL dataset is directly accessible via PhysioNet at <https://physionet.org/content/ptb-xl/1.0.3/>. The MIMIC-IV dataset is available to credentialed users through PhysioNet at <https://physionet.org/content/mimiciv/>, which requires completion of a data use agreement and human subjects training. Details on the credentialing process, which is open to all researchers, can be found at <https://physionet.org/about/citi-course/>. Due to MIMIC data protection policies, the authors cannot directly share the MIMIC-IV dataset. However, researchers who complete the credentialing process can access the dataset via the links above. Furthermore, by following the procedures described in Supplementary Text 3, they can extract the same subset of data used in this study. The data generated during this study are available from the authors upon reasonable request, for researchers who can provide proof of completing the credentialed access process (contact: LYLM85@gmail.com, jangood1122@gmail.com).

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Author contributions

J.-H.J. designed the study and drafted the main manuscript text. Y.-Y.J., S.K. and J.M.S. performed data curation and analysis. H.S.L. and J.-m.K. prepared the figures and contributed to interpreting the results. M.S.L., as corresponding author, conceived the overall concept, supervised the research, and finalized the manuscript. All authors reviewed and approved the final version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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